

Transfer Learning for Stock Returns Forecasting

Using Quasi-Randomized Networks with Synthetic Pretraining

Work in Progress

The Question

Traditional Approach to forecasting stock returns

- Fit ARCH models to forecast each stock separately
- Need to specify the type of ARCH (GARCH, APARCH, etc.) model

Experimental Transfer Learning Approach

- Pretrain a model on synthetic data distributed according to stock market stylized facts
 - Capture market dynamics
 - Apply the same model to any series of real-world stock returns
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Transfer Learning with ridge2f

Quasi-Randomized Networks for Time Series Forecasting

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Synthetic Data Generation

 **1,000 Paths × 500 Days**
Diverse synthetic market scenarios

 **Heston Model**
Stochastic volatility

 **Regime Switching**
Low/high volatility states

 **Jump Diffusion**
Sudden price shocks

 **Noise**
Market microstructure



2

Bayesian Optimization

(Doubly Constrained Nonlinear Ridge Regression)

 **Objective**
Minimize median RMSE across all series

 **Optimize**

- Quasi-random features
- Lookback window
- Regularization
- K-means centers

 **Save**
Pretrained checkpoint (.rds file)

 **Fast**
No backpropagation needed



3

Transfer to Real Stocks

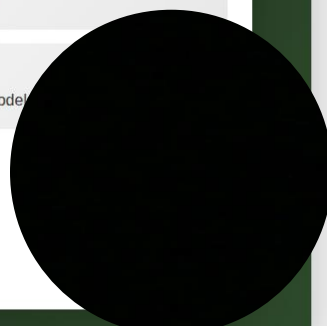
 **Frozen**
Hyperparameters from synthetic pretraining

 **Apply**
To real stock log-returns

 **Instant Fine-tuning**
Via **doubly constrained nonlinear ridge regression**

 **Deploy**
Zero per-stock tuning time

 **Results**
Competitive with GARCH models



Next Steps & Future Work

Current Status:

✓ Results are recent but robust across 43 stocks and multiple markets (on par with GARCH)

Future Work:

- The Why (it works)?
- Calibration via conformal prediction
- Parallelize the objective function
- Test on additional asset classes (crypto, e.g.)
- Benchmark against (fine-tuned) foundation models (TimeGPT, Chronos, etc.)



Resources

GitHub Repository

github.com/thierrymoudiki/2025-09-05-transfer-learning-ridge2f

Documentation & Papers:

- • ahead R package: docs.techtonique.net/ahead
- • ahead Python: docs.techtonique.net/ahead_python
- • Original paper on ridge2: MDPI Risks 2018
- • Blog posts: thierrymoudiki.github.io/blog

Questions?